

Sampling Conditions for Clipping-free Voronoi Meshing by the VoroCrust Algorithm

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Given a bounded open set $\mathcal{O}$ in Euclidean space, decompose its interior into Voronoi cells of bounded aspect-ratio. The cells should naturally *conform* to the bounding surface $\mathcal{M} = \partial\mathcal{O}$.

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- This is in contrast with tetrahedral meshing, which is well-established.
- The VoroCrust project developed the first solution to this problem.
- In this talk, we focus on the subproblem of surface reconstruction, assuming a set of sample points from $\mathcal{M}$ is given as input.

Surface Reconstruction for Voronoi Meshing

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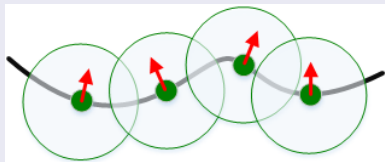
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 - Generators (poles) are restricted to lie near the medial axis of $\mathcal{M}$.
 - Instead of the Voronoi diagram, it is based on the power diagram.
 - Nonetheless, rich theory with strong approximation guarantees.

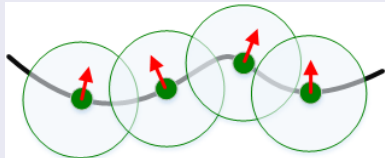
VoroCrust Intuition - A 2D Example

(1) Weighted samples (balls)

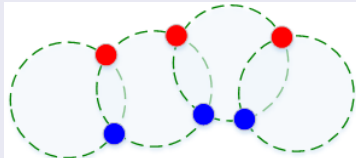


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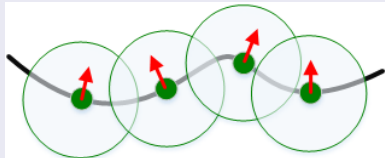


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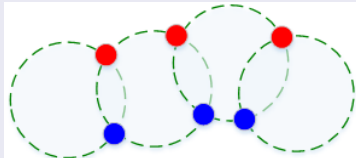


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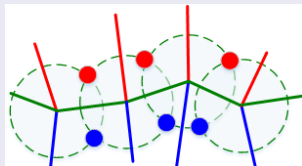
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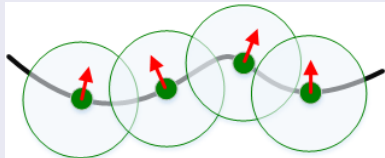


(3) Compute Voronoi diagram

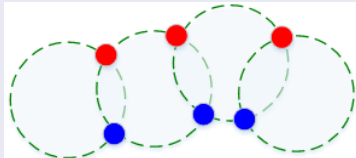


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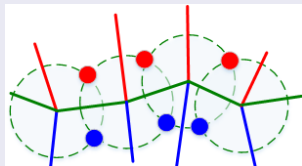
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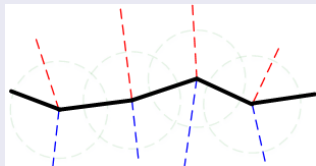
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(4) Keep the separating facets



The Union of Balls and Its Dual Shape [Edelsbrunner]

Power distance and cell

For a ball b centered at c with radius r , $\pi(b, x) = \|cx\|^2 - r^2$.

$V_b = \{x \in \mathbb{R}^d \mid \pi(b, x) \leq \pi(b', x) \forall b' \in B\}$.

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Define $\mathcal{K} = \text{Nerve}(\{V_b \cap b \mid b \in B\})$ and $\mathcal{S}$ as the underlying space $|\mathcal{K}|$.

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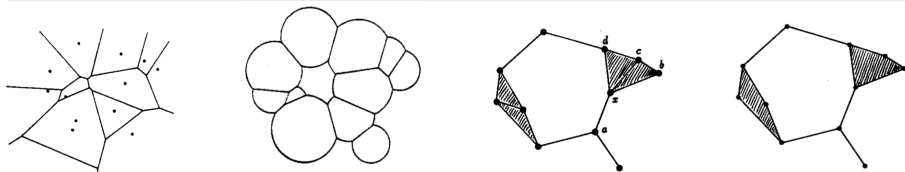
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Figures from [Edelsbrunner]

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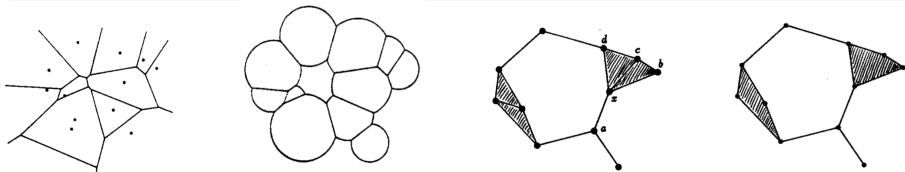
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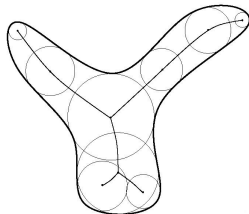
Homotopy-equivalence

The nerve theorem implies $\mathcal{S} = |\mathcal{K}|$ has the same homotopy-type as $\cup B$.

ϵ -Sampling and Topological Thickening

Local features size (lfs)

The local feature size at a point $x \in \mathcal{M}$ is its distance to the medial axis of $\mathcal{M}$.



Medial Axis
Figure from [Wolter]

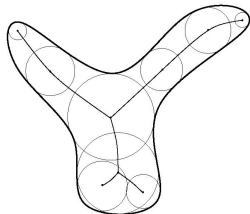
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A set of points P on $\mathcal{M}$ such that $\forall x \in \mathcal{M} \exists p \in P$ s.t. $\|px\| \leq \epsilon \cdot lfs(x)$.



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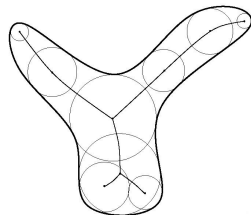
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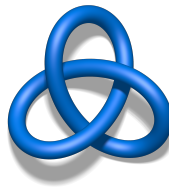
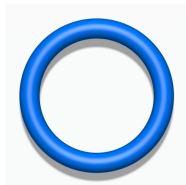


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From balls to surfaces [Chazal, Lieutier]

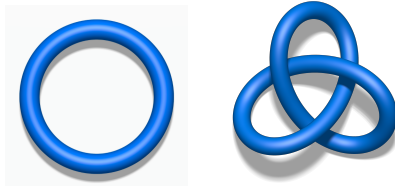
Let P be an ϵ -sample of $\mathcal{M}$, with $\epsilon < 1/160$, and define b_p as the ball centered at $p \in P$ with radius $\alpha_p \cdot lfs(p)$, where $1/20 < \alpha_p < 1/10$. Then, $\mathcal{M}$ is a deformation retraction of $\cup b_p$.

Beyond Homeomorphism: Isotopic Equivalence



Torus: unknot vs. knot
Figures from [Wikipedia]

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A purely topological condition [Chazal, Cohen-Steiner]

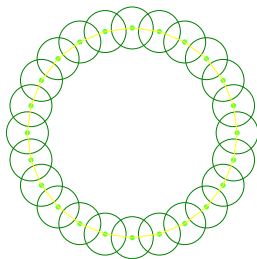
Suppose that:

- $\mathcal{M}'$ is homeomorphic to $\mathcal{M}$,
- $\mathcal{M}'$ is included in a topological thickening $\mathbb{M}$ of $\mathcal{M}$,
- $\mathcal{M}'$ separates the sides of $\mathbb{M}$.

Then, $\mathcal{M}'$ is isotopic to $\mathcal{M}$ in $\mathbb{M}$.

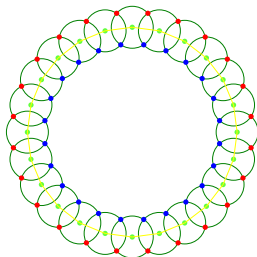
VoroCrust - The Abstract Algorithm

- Start with an ϵ -sample $P \subset \mathcal{M}$ with weights ($\delta \geq \epsilon$) defining the associated δ -lfs balls B .



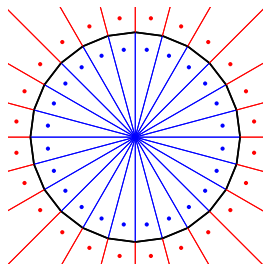
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- Start with an ϵ -sample $P \subset \mathcal{M}$ with weights ($\delta \geq \epsilon$) defining the associated δ -lfs balls B .
- Collect the corner points $\mathcal{G}$ of $\cup B$ as a crude set of witnesses of the simplices in $\mathcal{K}$.



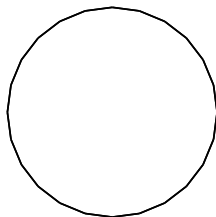
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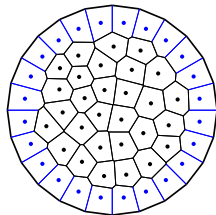
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- Produce the surface approximation as the facets of $Vor(\mathcal{G})$ separating interior/exterior corners; we call this the VoroCrust of $\mathcal{G}$ (VC).



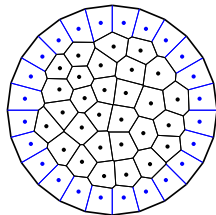
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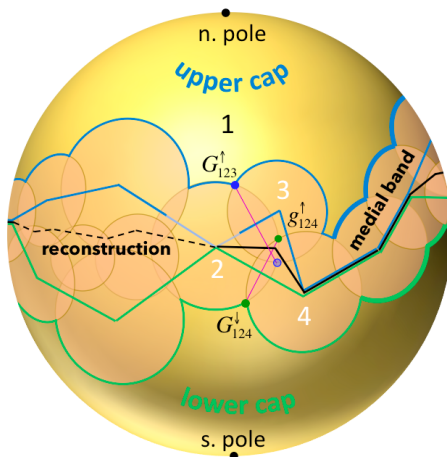
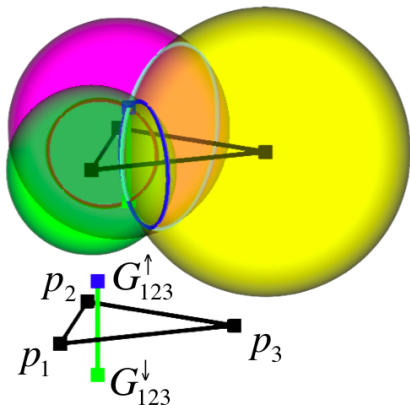
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Sandwich theorem

$$VC \subseteq \mathcal{S} \subseteq \cup B.$$

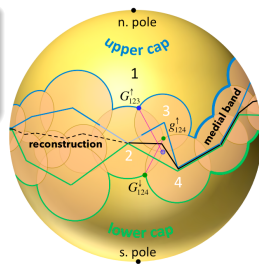
VoroCrust - Ball Intersections



VoroCrust - Sampling Conditions for Disk Caps

Disk Caps

Each sample ball contributes exactly two caps, i.e., topological-disks, to the boundary of the union.



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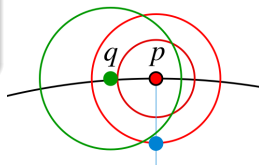
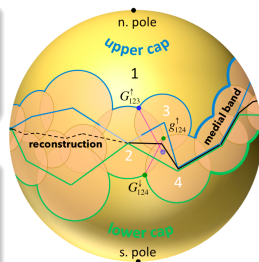
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Sampling Conditions

For constants $\epsilon \leq \delta$, we require an ϵ -sampling P , with associated balls of radii $r_i = \delta \cdot lfs(p_i)$ satisfying the following sparsity condition:

$$lfs(q) \geq lfs(p) \implies \|p - q\| \geq \epsilon \cdot lfs(p).$$



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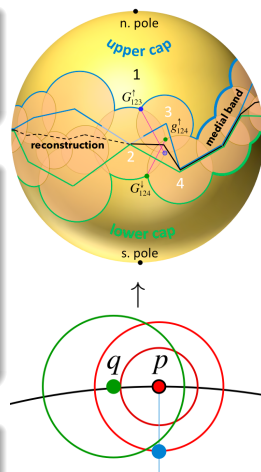
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Lemma

Taking $\epsilon = 1/160$ and $\delta = 1/20$, we get disk caps.



Announcements

- The VoroCrust software package is scheduled for release soon.
 - Successful implementation of more ideas than what this talk covers, e.g., sharp features, medial axis approximation, sizing estimation.
- Mohamed S. Ebeida (msebeid@sandia.gov) is the VoroCrust point-of-contact at Sandia National Labs (SNL).

